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DeepTrace Mobile: Viewer Forensic Cockpit

Architecture, UI/UX Page Flow, and Technical Engineering Specification for the Dedicated Mobile Evidence Review & Decision Platform

Document Purpose:	Mobile Engineering Specification & UI/UX Blueprint
Target Platform:	Cross-Platform Mobile Application (Flutter 3.x / Dart for Android & iOS)
Target User Persona:	Authorized Reviewers, Credit Risk Officers, Branch Heads & Legal Auditors
Security & Custody:	Read-Only Permission Scope, Ephemeral Decryption & Biometric Lock

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1. Executive Summary & Purpose

DeepTrace Mobile is engineered as a dedicated, cross-platform **Forensic Review & Decision Cockpit** designed specifically for the **Viewer / Reviewer persona**. While document ingestion, bulk queueing, and heavy multi-agent forensic computation are orchestrated by the high-concurrency FastAPI cloud backend and the desktop administrative portal, the mobile application empowers field risk officers, credit approval committees, branch heads, and judicial auditors to securely inspect forensic findings, interactively examine pixel-level evidence overlays, interrogate the multi-agent AI swarm, and review court-admissible audit reports directly from iOS and Android devices.

Architectural Mandate: The DeepTrace mobile application is purposefully restricted to a **Read-Only Forensic Viewer**. It eliminates camera-based intake, optical capture fragility, and document modification capabilities to preserve absolute cryptographic chain of custody under NIST SP 800-86 and Electronic Transactions Ordinance (ETO 2002) legal standards.

2. Strategic Rationale for a Dedicated Viewer-Only Mobile App

Restricting the mobile application exclusively to authorized forensic viewers delivers significant operational, security, and architectural advantages:

- **Operational Reality of Financial Institutions:** In commercial banks, micro-lenders, and visa agencies, physical documents are ingested via calibrated flatbed branch scanners or secure customer web portals. Executive decision-makers (Credit Approval Officers, Risk VPs, Branch Heads) do not scan paper on mobile phones—they require a responsive, high-fidelity mobile viewer to review flagged anomalies, inspect ELA heatmaps, and make time-sensitive credit or legal decisions on the go.
- **Preservation of Forensic Evidence Integrity:** Mobile camera captures introduce lighting glare, trapezoidal perspective distortions, sensor noise, and dynamic compression artifacts that can trigger false positives in delicate computer vision pipelines. By decoupling intake from review, DeepTrace ensures the forensic engines evaluate pristine digital artifacts while the mobile viewer delivers 100% reliable inspection.
- **Strict Role-Based Access Control (RBAC) & Immutability:** Under the State Bank of Pakistan (SBP) risk frameworks and PECA 2016, an investigator or reviewer must never possess client-side privileges to mutate, delete, or re-sign evidentiary files. A viewer-only app guarantees zero client-side data mutation risks, ensuring all evidence manifests remain cryptographically locked.
- **Optimized UI/UX Focused on Deep Explainability:** Instead of cluttering the application with camera controls, crop tools, and upload managers, 100% of the mobile interface is dedicated to rich forensic visualization: sub-pixel zoomable canvases, Error Level Analysis (ELA) toggles, font baseline grids, mathematical balance mismatch drawers, and conversational AI interrogation.

3. Security Architecture & Read-Only Governance

The mobile viewer interacts with the DeepTrace backend under an enforced **Zero-Trust Read-Only Contract**:

- **Hardware-Backed Biometric Lock:** App launches and sensitive case decrypts require local biometric validation (TouchID / FaceID / Android Biometrics) through the platform hardware security enclave.
- **Scoped JWT Tokens with Row-Level Security:** Viewer JWT tokens are strictly minted with role: VIEWER and scoped to the user's specific organizationId, backed by 19 PostgreSQL Row-Level Security policies.
- **Ephemeral Decryption & No Raw Storage:** High-resolution evidence tiles and bounding boxes are cached in encrypted memory-mapped storage and purged upon session logout or inactivity timeout.
- **Immutable Read Audit Trail:** Every document viewing, zoom action, layer toggle, and export event is logged back to the PostgreSQL audit_logs table with IP address, timestamp, and device fingerprint.

4. End-to-End Screen Navigation Architecture

The application follows a clean, single-purpose navigational hierarchy tailored for rapid forensic triage and in-depth verification:

Screen Name	Route / Path	Primary User Objective	Core Forensic Interactions
1. Biometric Auth	/auth	Authenticate viewer securely	Biometric prompt, hardware keystore unlock, tenant branch display.
2. Command Dashboard	/home	Triage urgent risk alerts	Critical case ticker, risk distribution dials, recent reviews feed, QR scanner.
3. Case Directory	/cases	Search & filter case files	Fuzzy search by Case ID/CNIC, filter by risk tier (Critical/High), bank name.
4. Quick QR Lookup	/lookup	Instant case retrieval	Scans QR code printed on audit certificate to pull case file in <1 second.
5. Case Verdict Hub	/cases/:id	Executive decision summary	0–100 Risk Dial, AI Lead Investigator verdict, Tri-Pillar indicator badges.
6. Forensic Viewer	/cases/:id/viewer	Visual evidence inspection	Pinch-to-zoom canvas, ELA heatmap switch, font baseline grid, anomaly tap.
7. Evidence Explorer	/cases/:id/evidence	Granular findings catalog	Deterministic math ledger checks, PK-IBAN check digits, font deviation scores.
8. AI Copilot Chat	/cases/:id/chat	Conversational interrogation	Natural language Q&A; bound to verified evidence, clickable canvas jump-links.
9. Audit Report	/cases/:id/report	Legal compliance review	SHA-256 custody verification, ETO 2002/PECA checklist, court-admissible PDF share.
10. Settings & Vault	/settings	Device security & cache	Offline encrypted case cache management, session timeout, SBP legal references.

5. Detailed Screen Specifications (Discovery & Access)

Screen 1: Biometric Authentication Screen (/auth)

Serves as the zero-knowledge security gateway. When opened, the app queries the platform hardware keystore. Upon successful fingerprint or FaceID verification, the device's securely stored refresh token exchanges for an ephemeral session token. UI displays the viewer's institutional branch affiliation (*e.g., Habib Bank Limited — Head Office Risk Division*), their authenticated role (AUDITOR / REVIEWER), and an active SBP compliance badge.

Screen 2: Viewer Command Dashboard (/home)

Designed for rapid triage by risk managers. Features a top KPI carousel displaying: (1) *Critical Cases Awaiting Adjudication* (colored in crimson), (2) *High Suspicion Documents* (amber), and (3) *Clean Verifications* (green). Includes a prominent Quick Case Lookup field supporting instantaneous regex parsing of case numbers (DT-PK-YYYY-NNNNNN), and a Quick QR Scan button that activates a lightweight camera overlay strictly to read forensic verification QR seals on printed paper certificates.

Screen 3: Case Directory & Advanced Forensic Search (/cases)

Provides an exhaustive, paginated repository of processed cases with real-time trigram search capabilities. Viewers can filter across: (a) Document category (*Bank Statements*, *Salary Slips*, *FBR Tax Withholding Certificates*, *Utility Bills*), (b) Issuing banking institution (*HBL*, *Meezan*, *MCB*, *Alfalalah*), (c) Risk tier score sliders (0–100), and (d) Pipeline analysis date. Case preview cards highlight the total detected anomaly count, initial SHA-256 hash preview, and time elapsed since processing.

6. Flagship Screen: Interactive Forensic Document Viewer (/cases/:id/viewer)

The **Interactive Forensic Viewer** represents the core technical achievement of the DeepTrace mobile application. Engineered using Flutter's high-performance InteractiveViewer and hardware-accelerated CustomPainter canvas, it enables seamless sub-pixel pinch-to-zoom (up to 800% magnification) and continuous 60fps panning over multi-page high-resolution document scans.

Core Innovation — Dynamic Multi-Layer Forensic Stack: Rather than displaying a static rendered image, the viewer maintains a hardware-accelerated composite stack where the viewer can instantaneously toggle between five specialized forensic observation layers.

- **Layer 1: Pristine Original Scan:** Displays the unmodified high-resolution document scan or vector PDF rendering in its true original DPI without any graphic annotations, serving as the visual baseline.
- **Layer 2: Error Level Analysis (ELA at Q=95) Heatmap:** Renders the false-color compression gradient generated by DeepTrace's OpenCV forensic engine. Unmodified text areas appear dark purple/black, while re-saved, edited, or spliced regions (such as pasted numbers, stamps, or altered figures) glow brightly in vivid white, cyan, and magenta.
- **Layer 3: Vector Bounding Box Overlay:** Overlays precision vector bounding boxes around all flagged anomalies. Critical anomalies (e.g., balance calculation discrepancies, copy-move forgery clusters) are bordered in red, while moderate visual anomalies (e.g., minor font baseline shifts) are framed in amber. Each box features an interactive numbered badge.
- **Layer 4: Sub-Pixel Font Baseline & Glyph Grid:** Projects horizontal sub-pixel alignment lines across detected text rows. Characters or numbers that have been spliced or typed using third-party desktop editors display visible vertical displacement above or below the true typographic baseline.
- **Layer 5: Copy-Move Forgery Clustered Regions:** Visualizes duplicated image patches or cloned text blocks using connected bezier arcs between the source region and the destination clone, immediately exposing copied bank stamps or duplicated transaction rows.

Interactive Anomaly Inspection Drawer & Comparison Slider

1. Tap-to-Inspect Anomaly Drawer: When a viewer taps on any highlighted bounding box on the document canvas, a bottom sheet modal slides up with detailed forensic metadata: (a) *Anomaly Classification* (e.g. *Copy-Move Forgery* or *Math Balance Discrepancy*), (b) *Confidence Level* (*98.2%*), (c) *Forensic Engine Rule* (*ELA Gradient Discontinuity + SBP Balance Reconciliation Failure*), and (d) *Exact Pixel Bounding Box Coordinates* [x: 412, y: 188, w: 94, h: 22].

2. Real-Time Comparison Diff Slider: The viewer includes a draggable vertical comparison divider. Sliding it horizontally across the document splits the screen, showing the pristine document on the left and the active forensic ELA heatmap on the right, giving the reviewer undeniable proof of manipulation.

7. Case Verdict & Risk Overview Hub (/cases/:id)

Serves as the decision-making landing screen before deep visual inspection. It aggregates the multi-agent findings into a clear verdict:

- **Radial Risk Gauge (0–100):** Displays an animated speedometer-style gauge calibrated into four risk bands: 0–25 LOW (Green), 26–60 MEDIUM (Yellow), 61–85 HIGH (Orange), and 86–100 CRITICAL (Red).
- **Executive Verdict Card:** Generates a natural-language executive summary from the Lead Investigator Agent: *'Document exhibits high probability of digital tampering. 3 critical visual splices and 1 mathematical balance contradiction detected.'*
- **Tri-Pillar Findings Summary:** Summarizes the three forensic pillars (Deterministic, Computer Vision, Semantic PK) with pass/fail indicator chips.
- **Direct Navigation Controls:** Prominent action buttons to launch the *Interactive Evidence Viewer*, *Ask DeepTrace Chat*, or *Export Audit Report*.

8. 'Ask DeepTrace' Forensic Chat Copilot (/cases/:id/chat)

A standout capability of DeepTrace is its **Conversational Forensic Copilot**. Rather than leaving risk officers to decipher raw cryptographic data, viewers can interrogate the case conversationally in natural language.

Strict Zero-Hallucination Policy: The mobile chat interface communicates with the LangGraph Multi-Agent Swarm operating under strict validation gates. The model is forbidden from answering with generic speculation; every response must be directly bound to verified evidence items stored in the PostgreSQL database.

- **One-Tap Reviewer Quick Prompts:** Provides smart chips tailored for banking review: *'Why is this document flagged Critical?'*, *'Show me all mathematical calculation errors.'*, *'Is there evidence of Photoshop software editing?'*, and *'Does the account name match FBR tax records?'*
- **Interactive Evidence Jump-Links:** When the AI explains an anomaly (e.g. *'Transaction #4 on June 12 was spliced'*), the response includes a clickable inline pill: [View Anomaly #3]. Tapping this citation automatically switches to the Interactive Document Viewer, centers the canvas on that exact bounding box, and pulses the highlight.
- **Bilingual Urdu/English Support:** To support field investigators across Pakistani regional branches, the chat engine supports both English and Urdu queries and responses.
- **Evidence Pinning & Annotation:** Reviewers can bookmark key AI findings directly into the case adjudication notes to be appended to the final institutional report.

9. Tri-Pillar Evidence Explorer (/cases/:id/evidence)

Provides a comprehensive, tabular catalog of all individual evidence items discovered during pipeline execution, structured into three distinct tabs:

- **Tab 1 — Deterministic Financial & Structure Checks:** Lists mathematical ledger reconciliations (calculating credit/debit balances row by row), verifies Pakistani IBAN MOD-97 check digits, checks CNIC checksum algorithms, and inspects PDF object trees for incremental modification saves.
- **Tab 2 — Computer Vision & Compression Forensics:** Displays quantitative metrics from Error Level Analysis (peak compression divergence at Q=95), sub-pixel font baseline displacement values in points, and copy-move block similarity matrices.
- **Tab 3 — Multi-Agent Reasoning & Semantic Manifests:** Presents individual diagnostic logs from the Structural Agent, Visual Forensics Agent, and Semantic Pakistan Agent, showing the chain-of-thought evidence fusion.

10. Forensic Audit Report & Chain of Custody (/cases/:id/report)

For legal proceedings and formal bank risk committee approvals, this screen renders a court-admissible audit certificate including: (1) *Cryptographic Custody Manifest* showing initial SHA-256 ingestion hash, processing node ID, and timestamp; (2) *Statutory Compliance Checklist* validating conformity with the Electronic Transactions Ordinance (ETO 2002) § 3 & 4 and PECA 2016 § 32; (3) *Adjudication Sign-Off Block* allowing the viewer to record an institutional verdict (*Approved / Rejected / Escalated to FIA Cybercrime*); and (4) *Export & Share* buttons to generate an official watermarked PDF report with an embedded verification QR seal.

11. Technical Implementation Stack & Flutter Architecture

The mobile application is developed following clean feature-first architecture principles in Flutter 3.x:

Component Layer	Selected Technology	Technical Role & Justification
Cross-Platform UI	Flutter 3.x / Dart 3.x	Delivers 60fps hardware-accelerated canvas rendering across Android and iOS from a unified codebase.
State Management	Riverpod 2.x	Compile-safe, reactive state architecture managing auth tokens, case filters, and multi-layer viewer states.
Document Canvas	InteractiveViewer + CustomPainter	Handles smooth 2D transformations (pan, scale, matrix operations) with custom vector bounding box rendering.
Network & Streaming	Dio + web_socket_channel	High-performance HTTP client with interceptors for JWT injection and WebSocket streaming for live telemetry.
Secure Storage	flutter_secure_storage	Stores JWT session tokens and encryption keys inside iOS Keychain and Android Keystore.
Offline Cache Vault	Hive (Encrypted Box)	Ultra-fast NoSQL local database storing downloaded evidence manifests for offline viewing.

12. REST API & WebSocket Communication Protocol

The viewer client consumes the following specialized endpoints exposed by the DeepTrace FastAPI backend:

Method & Endpoint	Purpose & Data Payload
POST /api/v1/auth/login	Authenticates viewer with biometric hash / password; returns JWT with VIEWER claim.
GET /api/v1/investigations	Retrieves paginated case archive with filters: status, riskTier, orgId.
GET /api/v1/investigations/{id}	Returns complete case manifest: risk score (0–100), AI verdict, and document metadata.
GET /api/v1/investigations/{id}/evidence	Delivers all bounding boxes, ELA heatmap asset URLs, and deterministic math discrepancies.
POST /api/v1/investigations/{id}/ask	Submits natural language queries to the LangGraph AI swarm; returns verified citations.
GET /api/v1/investigations/{id}/report	Downloads cryptographically sealed, court-admissible forensic PDF report with QR stamp.

13. Development Roadmap (Samreen Aslam's Sprint Plan)

Sprint	Timeline	Milestones & Core Deliverables
Sprint 1	Weeks 1–2	Project scaffolding, clean architecture setup, Riverpod providers, Biometric Auth integration.
Sprint 2	Weeks 3–4	Viewer Command Dashboard, Case Directory UI, fuzzy search & advanced filtering with FastAPI API.
Sprint 3	Weeks 5–7	Interactive Forensic Viewer: High-DPI canvas, matrix pinch-zoom, ELA layer switcher & bounding boxes.
Sprint 4	Weeks 8–9	Comparison slider diffing, tappable anomaly inspection bottom sheets, Tri-Pillar Evidence Explorer.
Sprint 5	Weeks 10–11	Conversational AI Copilot Chat ('Ask DeepTrace') with interactive canvas jump-links, QR code lookup.
Sprint 6	Week 12	Audit report PDF export, offline encrypted Hive cache, end-to-end UAT testing, final submission.